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Cartesian trajectory planning of space robots using a multi-objective optimization

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ABSTRACT

Cartesian trajectory planning of a free-floating space robot is impacted by dynamic singularities due to the inverse kinematics equations. Although various methods have been proposed to avoid the singularities, very few of them are suitable for the trajectory planning of a high degree-of-freedom (DOF) space robot in the Cartesian space. In this paper, a method of combining Damped Least Squares (DLS) and feedback compensation is developed to avoid such singularities. The trajectories of the end-effector are parametrized with Bézier curves, which are simple and make it easy to limit the joint velocities. Moreover, because of certain missions, such as communication and observation, base attitude disturbance and moving time are considered to establish a cost function and the trajectory planning is transformed into a multi-objective optimization. Chaotic particle swarm optimization (CPSO) is employed to solve the optimization, which can improve the premature phenomenon of particle swarm optimization (PSO). Simulation results are presented for the trajectory planning of a 6 DOF space robot and demonstrate the effectiveness of the proposed method.

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1. Introduction

With the rapid development of space technology, space robots play a greater role in the exploration of space resources. Examples are the ETS-VII [1] and TianGong-2 robotic hand [2]. They can complete certain tasks in harsh space environment instead of humans [1–3].

To conserve fuel or avoid interactions with nearby objects, all spacecraft actuators may be turned off. Then a space robot operates in a so-called free-floating mode, which is of great interest to researchers [4–6]. Trajectory planning is particularly important for applications of free-floating space robots. Similar to trajectory planning of ordinary robots, it involves the singularity avoidance, the limitation of the joints, and the optimization of energy or time. However, in this mode, the position and the attitude of the spacecraft (the base) are not controlled, and the spacecraft moves as a consequence of manipulator motions, due to the dynamic coupling. Moreover, suppose that there are no external forces or torques. Because of nonholonomic constraints (angular momentum conservation), the state of the end-effector depends not only on the joint variables but also on the time history of their trajectories. Therefore, trajectory planning of space robots is more complex.

For Cartesian trajectory planning of space robots, when the generalized Jacobian matrix is not invertible, dynamic singularities occur. Many research results deal with dynamic singularities. To avoid the singularities, trajectories of joint angles, joint velocities and joint accelerations, were parametrized in [7–10]. Since only the forward kinematic equation was employed, the methods of parametrizing trajectories in joint space are not affected by dynamic singularities. However, they are obviously not suitable for Cartesian trajectory planning [11]. Papadopoulos [12] proposed a method of avoiding dynamic singularities based on the reachable workspace and applied it to a 2 DOF space robot. For a high DOF space robot, it was much more difficult to obtain the workspace. In terms of a space robot with a PUMA-type manipulator, a method of combining the singularity separation and damped reciprocal was developed in [13]. The dynamic singularities were avoided by adjusting the initial configuration of the space robot including the joint angles and the spacecraft attitude [11,14]. Therefore, base attitude disturbance was ignored, which was not expected in some missions such as communicating with the earth. Damped Least Squares (DLS) is a common way to deal with the singularities for an ordinary robot [15,16]. It could also be applied to space robots, however, it might lead to trajectory errors. Therefore, there are very few dynamic-singularity-avoidance methods, which

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are suitable for Cartesian trajectory planning of a high DOF space robot.

Large attitude disturbance to the base is not expected during the motion of the space manipulator in certain missions such as observing the earth. There are actually some results about reducing the base attitude disturbance. Yoshida [17] investigated Reaction Null-Space (RNS) and the base attitude was not influenced by the motion of the manipulator. Trajectory planning algorithms combining RNS and optimal methods were developed [9,18]. Xu [19] proposed adaptive zero reaction motion control. Although RNS can guarantee that the base attitude disturbance is zero, it is not suitable for a nonredundant system. Moreover, kinematics and dynamics of redundant space robots are much more complex [20]. Space robots with dual manipulators were studied in [21]. One manipulator was used to finish the missions and the other was used to compensate the base attitude disturbance. The dynamics and kinematics of dual manipulators are also complex. Based on a novel hybrid dynamic coupling model, the joint-to-base and end-to-base coupling maps were established to minimize the base attitude disturbance [22]. To reduce the disturbance to the base attitude, some researchers employed optimization methods. The disturbance was regarded as a part of the cost function and the optimizations were solved with particle swarm optimization (PSO) [23]. Misra [10] transformed the trajectory planning of space robot to a convex quadratic programming. Both Huang [24] and Liu [25] considered the rejection of the disturbance as a corresponding constraint in the optimizations.

The trajectory planning is also transformed into optimizations for other objectives. In point-to-point trajectory planning, to guarantee the accuracy of the end-effector trajectory at the final moment, some researchers applied optimization methods such as the differential evolution algorithm [10], PSO [8,18,23,26] and the genetic algorithm [27]. In addition, some researchers have studied the trajectory planning considering optimal time or energy. Huang [24] developed a time-optimized trajectory planning algorithm for space robots. Time-optimal trajectory planning methods have also been developed for delta robots, mobile robots, and spherical robots [28–30]. In [25], trajectories of space robot with optimal energy were obtained based on Gauss pseudo-spectral method. Some researchers focused on trajectory planning based on multi-objective method. Path length, smoothness, and safety were combined to form a multi-objective path planning problem for multiple mobile robots. In [31], the multi-objective optimization was employed to obtain the trajectory considering both time and energy for flexible space robots.

Based on the above, there are abundant research results in trajectory planning with consideration of dynamic singularities. However, in terms of Cartesian trajectory planning of a high DOF space robot, there are very few dynamic-singularity-avoidance methods. Moreover, in certain missions, the base attitude disturbance and the time to complete the movement have to be minimized. Therefore, for a 6 DOF space robot, there are much fewer Cartesian trajectory planning methods, which consider dynamic singularities, the base attitude disturbance, and the moving time together. This study will provide a new possible solution to this problem. The main contributions and differences of this study relative to existing works are as follows:

1. A method of combining DLS and feedback compensation is proposed to avoid dynamic singularities. DLS causes that the actual trajectories deviate from the desired trajectories. Different from DLS, the proposed method can gradually reduce or even eliminate the error by introducing feedback compensation.
2. Considering the base attitude disturbance, the moving time and the joint-angle requirements, a cost function is established and the trajectory planning is transformed into a multi-objective optimization problem. CPSO is adopted to solve the optimization, which can overcome the premature phenomenon of PSO effectively.
3. The trajectories of the end-effector are parametrized with Bézier curves, which can make it easy to limit the joint velocities. Compared with polynomial functions, Bézier curves can simplify the calculation.

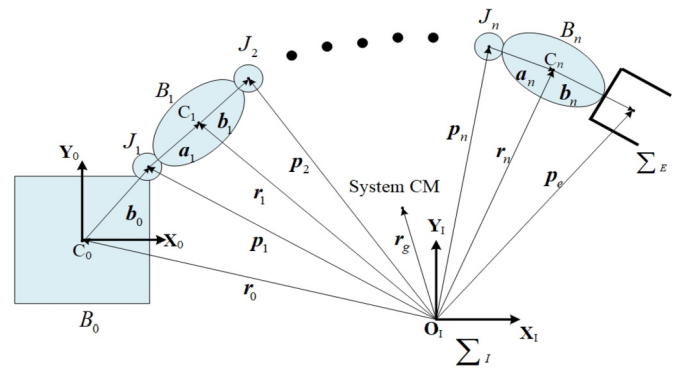

Fig. 1. Schematic diagram of a space robot.

Table 1
Kinematic and dynamic symbols of a space robot.

Symbols	Representation
B_0, B_i	The base and the i-th link of the manipulator ($i = 1, 2, \cdots, n$)
C_0, C_i	Mass centers of B_0 and B_i
J_i	A point in the i-th joint, connecting B_{i-1} and B_i
$\boldsymbol{a}_i, \boldsymbol{b}_i$	Position vectors from J_i to C_i and C_i to J_{i+1}
$m_i, \boldsymbol{I}_i$	Mass and inertia matrix of B_i
$\boldsymbol{r}_i, \boldsymbol{r}_g$	Position vectors of C_i and system mass center
$\boldsymbol{p}_i, \boldsymbol{p}_e$	Position vectors of J_i and the end-effector
$\boldsymbol{\omega}_e, \boldsymbol{\omega}_0$	Angular velocities of the end-effector and B_0
$\boldsymbol{\Theta} = [\theta_1, \theta_2, \ldots, \theta_n]^{\mathrm{T}}$	The vector of joint angles of the manipulator

The rest of this paper is organized as follows. In Section 2, kinematics of a free-floating space robot is introduced. In Section 3, the trajectory planning, considering dynamic singularities, the base attitude disturbance and the moving time, is transformed into a multi-objective optimization. CPSO algorithm is presented in Section 4. The simulation results of the developed trajectory planning algorithm are shown in Section 5. Finally, conclusions based on the simulation results are given in Section 6.

2. Modeling of free-floating space robot

2.1. Kinematics of space robot

A space robotic system consists of a spacecraft (base) and a manipulator with n DOF as shown in Fig. 1. Kinematic and dynamic symbols of a space robot are listed in Table 1.

The kinematic mapping of the end-effector can be expressed as follows

$$\begin{bmatrix} \dot{\boldsymbol{p}}_e \\ \boldsymbol{\omega}_e \end{bmatrix} = \boldsymbol{J}_b \begin{bmatrix} \dot{\boldsymbol{r}}_0 \\ \boldsymbol{\omega}_0 \end{bmatrix} + \boldsymbol{J}_m \dot{\boldsymbol{\Theta}} \tag{1}$$

where $\boldsymbol{J}_b$ and $\boldsymbol{J}_m$ are Jacobian matrices of the base and the manipulator respectively, given in [32].

If there are no external forces and torques applied on the base, then the system is called a free-floating space robot. Therefore, momentum and angular momentum are conserved in the free-floating mode, which can be written as

$$\begin{bmatrix} \boldsymbol{P} \\ \boldsymbol{L} \end{bmatrix} = \boldsymbol{H}_b \begin{bmatrix} \dot{\boldsymbol{r}}_0 \\ \boldsymbol{\omega}_0 \end{bmatrix} + \boldsymbol{H}_{bm}\dot{\boldsymbol{\Theta}} \tag{2}$$

where $\boldsymbol{H}_b$ and $\boldsymbol{H}_{bm}$ are inertia matrix and coupling inertia matrix respectively, obtained in [32]. Suppose the initial values of $\boldsymbol{P}$ and $\boldsymbol{L}$ are zero. Then the motion of the base can be represented by

$$\begin{bmatrix} \dot{\boldsymbol{r}}_0 \\ \boldsymbol{\omega}_0 \end{bmatrix} = \boldsymbol{H}_b^{-1}\boldsymbol{H}_{bm}\dot{\boldsymbol{\Theta}} = \begin{bmatrix} \boldsymbol{J}_{bmv} \\ \boldsymbol{J}_{bm\omega} \end{bmatrix}\dot{\boldsymbol{\Theta}} = \boldsymbol{J}_{bm}\dot{\boldsymbol{\Theta}} \tag{3}$$

Substituting Eq. (3) into Eq. (1), the kinematic equations can be rewritten as

$$\begin{bmatrix} \dot{\boldsymbol{p}}_e \\ \boldsymbol{\omega}_e \end{bmatrix} = (\boldsymbol{J}_b\boldsymbol{J}_{bm} + \boldsymbol{J}_m)\dot{\boldsymbol{\Theta}} = \boldsymbol{J}_g\dot{\boldsymbol{\Theta}} \tag{4}$$

where $\boldsymbol{J}_g$ is termed Generalized Jacobian Matrix. Matrix $\boldsymbol{J}_g$ is related not only to the kinematic parameters, but also to the dynamic parameters [7]. Therefore, singularities of a free-floating space robot are called dynamic singularities.

2.2. Attitude representations of a rigid body

Euler angle is a common attitude representation of a rigid body. In this paper, the sequence of rotations $\mathrm{Z}(\alpha) - \mathrm{Y}(\beta) - \mathrm{X}(\gamma)$ is employed, and the attitude is described by Euler angles $\boldsymbol{\Phi} = [\alpha, \beta, \gamma]$. $\mathrm{Z}(\alpha)$, $\mathrm{Y}(\beta)$, and $\mathrm{X}(\gamma)$ denote the rotations of α, β, and γ about $\mathrm{Z}-$axis, $\mathrm{Y}-$axis, and $\mathrm{X}-$axis, respectively. The relation between the time derivative of the Euler angles $\boldsymbol{\Phi}$ (that we will call rotational velocity) and angular velocity $\boldsymbol{\omega}$ is

$$\boldsymbol{\omega} = \begin{bmatrix} 0 & -\sin\alpha & \cos\alpha\cos\beta \\ 0 & \cos\alpha & \sin\alpha\cos\beta \\ 1 & 0 & -\sin\beta \end{bmatrix} \begin{bmatrix} \dot{\alpha} \\ \dot{\beta} \\ \dot{\gamma} \end{bmatrix} = \boldsymbol{N}_\phi\dot{\boldsymbol{\Phi}} \tag{5}$$

Therefore, when $\cos\beta = 0$, the matrix $\boldsymbol{N}_\phi$ is not invertible. It means that the rotational velocity $\dot{\boldsymbol{\Phi}}$ cannot be expressed by means of $\boldsymbol{\omega}$. This phenomenon is called representation singularity of Euler angles.

Unit quaternion is an alternative attitude representation, particularly convenient because the relations are algebraic rather than trigonometric. A unit quaternion can be defined as

$$\boldsymbol{Q} = [q_0\, q_1\, q_2\, q_3]^{\mathrm{T}} = \begin{bmatrix} q_0 & \boldsymbol{q}^{\mathrm{T}} \end{bmatrix}^{\mathrm{T}} = \begin{bmatrix} \cos\dfrac{\psi}{2} & \sin\dfrac{\psi}{2}\boldsymbol{e}^{\mathrm{T}} \end{bmatrix}^{\mathrm{T}} \tag{6}$$

where $\boldsymbol{e}$ is the unit rotation axis and ψ is the rotation angle along $\boldsymbol{e}$. The quaternion satisfies $q_0^2 + \boldsymbol{q}^{\mathrm{T}}\boldsymbol{q} = 1$. Note that an attitude can be represented by both $\boldsymbol{Q}$ and $-\boldsymbol{Q}$. However, if $-\pi < \psi \le \pi\,(rad)$ and $q_0 \ge 0$ hold, the attitude expressed by unit quaternion is unique. The time derivative of the quaternion $\boldsymbol{Q}$ can be written as

$$\begin{bmatrix} \dot{q}_0 \\ \dot{\boldsymbol{q}} \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -\boldsymbol{q}^{\mathrm{T}} \\ q_0\boldsymbol{E} - \boldsymbol{q}^{\times} \end{bmatrix}\boldsymbol{\omega} \tag{7}$$

where $\boldsymbol{E}_n$ is an identity matrix with size $n \times n$, and $\boldsymbol{q}^{\times}$ is defined as

$$\boldsymbol{q}^{\times} = \begin{bmatrix} 0 & -q_3 & q_2 \\ q_3 & 0 & -q_1 \\ -q_2 & q_1 & 0 \end{bmatrix}$$

$\boldsymbol{\omega}$ can also be represented by $\boldsymbol{Q}$:

$$\boldsymbol{\omega} = 2\left(\frac{q_0^2\boldsymbol{E}_3 - q_0\boldsymbol{q}^{\times} + \boldsymbol{q}\boldsymbol{q}^{\mathrm{T}}}{q_0}\right)\dot{\boldsymbol{q}} \tag{8}$$

Remark 1. On the one hand, when $\cos\beta = 0$, the rotational velocity $\dot{\boldsymbol{\Phi}}$ cannot be expressed by means of $\boldsymbol{\omega}$. Therefore, if the attitude information is needed based on $\boldsymbol{\omega}$, it is more suitable to employ unit quaternion. On the other hand, when $q_0 = 0$, $\boldsymbol{\omega}$ cannot be expressed by means of $\boldsymbol{Q}$ according to (8). Thus, if $\boldsymbol{\omega}$ has to be solved based on the attitude information, it is better to employ the Euler angles.

3. Trajectory planning of space robot

Cartesian trajectory planning for a space robot is studied in this section. The objective of trajectory planning is to make the end-effector move continuously and smoothly from an initial state to a desired state. As already mentioned, due to the inverse kinematics equations, dynamic singularities should be considered. The Damped Least Squares (DLS) method is employed to avoid the singularities. However, this leads to an error between the desired trajectory and the actual trajectory. To overcome this drawback, a method of feedback compensation is introduced to reduce the error gradually. Moreover, the base attitude disturbance and moving time are considered as two parts of a cost function. Therefore, the trajectory planning is transformed into a multi-objective optimization.

3.1. Parametrization of the trajectories

In this section, we assume that the end-effector is desired to move along a straight line. The initial and final desired positions of the end-effector are $\boldsymbol{p}_{e0}$ and $\boldsymbol{p}_{ed}$, respectively. Then, the position and the velocity of the end-effector can be represented by

$$\boldsymbol{p}_e(t) = \boldsymbol{p}_{e0} + s_p(t)(\boldsymbol{p}_{ed} - \boldsymbol{p}_{e0}) \tag{9}$$

$$\dot{\boldsymbol{p}}_e(t) = \dot{s}_p(t)(\boldsymbol{p}_{ed} - \boldsymbol{p}_{e0}) \tag{10}$$

If the scalar time law $s_p(t)$ is determined, the trajectory of the position is determined.

The initial and final desired attitudes of the end-effector are $\boldsymbol{\Phi}_{e0} = [\alpha_{e0}, \beta_{e0}, \gamma_{e0}]^{\mathrm{T}}$ and $\boldsymbol{\Phi}_{ed} = [\alpha_{ed}, \beta_{ed}, \gamma_{ed}]^{\mathrm{T}}$ (ZYX Euler angles), respectively. Similarly, the Euler angles and angular velocities can be designed as

$$\boldsymbol{\Phi}_e(t) = \boldsymbol{\Phi}_{e0} + s_\Phi(t)(\boldsymbol{\Phi}_{ed} - \boldsymbol{\Phi}_{e0}) \tag{11}$$

$$\dot{\boldsymbol{\Phi}}_e(t) = \dot{s}_\Phi(t)(\boldsymbol{\Phi}_{ed} - \boldsymbol{\Phi}_{e0}) \tag{12}$$

For their simplicity and normalization, Bézier curves are widely used. Here, $s_p(t)$ and $s_\Phi(t)$ are parameterized with Bézier curves, which are also used in order to limit the angular velocities of the joints. If we define $\tau = t/T_f$ and $\tau \in [0\,1]$, $s_p(t)$ and $s_\Phi(t)$ can be expressed as follows

$$\begin{aligned} s_p(\tau) &= \textstyle\sum_{j=0}^{m} C_j^m (1-\tau)^{m-j}\tau^j P_{p\,j} \\ s_\Phi(\tau) &= \textstyle\sum_{j=0}^{m} C_j^m (1-\tau)^{m-j}\tau^j P_{\Phi\,j} \end{aligned} \tag{13}$$

For any variable χ, χ' represents the derivative of χ with respect to τ. s_p' and s_Φ' can be calculated as follows

$$\begin{aligned} s_p' &= \tfrac{\mathrm{d}s_p}{\mathrm{d}\tau} = m\textstyle\sum_{j=0}^{m-1} C_{m-1}^j (1-\tau)^{m-1-j}\tau^j (P_{p\,(j+1)} - P_{p\,j}) \\ &= T_f\dot{s}_p \\ s_\Phi' &= \tfrac{\mathrm{d}s_\Phi}{\mathrm{d}\tau} = m\textstyle\sum_{j=0}^{m-1} C_{m-1}^j (1-\tau)^{m-1-j}\tau^j (P_{\Phi\,(j+1)} - P_{\Phi\,j}) \\ &= T_f\dot{s}_\Phi \end{aligned} \tag{14}$$

Based on the initial and desired states of the end-effector, $s_p(\tau)$ and $s_\Phi(\tau)$ satisfy

$$\begin{aligned} s_p(0) &= s_\Phi(0) = 0 \\ s_p(1) &= s_\Phi(1) = 1 \end{aligned} \tag{15}$$

To move smoothly at initial and final moment, $s'_p(\tau)$ and $s'_\Phi(\tau)$ satisfy

$$s'_p(0) = s'_\Phi(0) = s'_p(1) = s'_\Phi(1) = 0 \tag{16}$$

According to Eqs. (13)-(16), the following equality constraints are obtained.

$$\begin{aligned} P_{p\,0} &= P_{p\,1} = P_{\Phi\,0} = P_{\Phi\,1} = 0 \\ P_{p\,(m-1)} &= P_{p\,m} = P_{\Phi\,(m-1)} = P_{\Phi\,m} = 1 \end{aligned} \tag{17}$$

Therefore, according to Eq. (17), the desired trajectories of the end-effector depend on $P_{p\,i}$ and $P_{\Phi\,i}$, $i = 2, 3, \ldots, (m-2)$. In this paper, $m = 4$ is supposed. Then, if $P_{p\,2}$ and $P_{\Phi\,2}$ are determined, the desired trajectories (9) and (11) are obtained.

Remark 2. If $m = 3$, the desired trajectories are obtained according to (17). Therefore, in order to achieve additional objectives such as reducing base attitude disturbance, $m \geq 4$ should be satisfied.

Define the end-effector position and attitude as $\boldsymbol{x}_e = \left[\boldsymbol{p}_e^{\mathrm{T}}\ \boldsymbol{\Phi}_e^{\mathrm{T}}\right]^{\mathrm{T}}$. Using the inverse kinematics equations, the joint velocities with respect to τ can be calculated as follows

$$\begin{aligned} \boldsymbol{\Theta}' &= \tfrac{\mathrm{d}\boldsymbol{\Theta}}{\mathrm{d}\tau} = \boldsymbol{J}_g^{-1}\begin{bmatrix} \boldsymbol{p}'_e \\ \boldsymbol{\omega}'_e \end{bmatrix} = \boldsymbol{J}_g^{-1}\boldsymbol{J}_{\Phi\omega}\begin{bmatrix} s'_p(\boldsymbol{p}_{ed} - \boldsymbol{p}_{e0}) \\ s'_\Phi(\boldsymbol{\Phi}_{ed} - \boldsymbol{\Phi}_{e0}) \end{bmatrix} \\ &= \boldsymbol{J}_g^{-1}\boldsymbol{J}_{\Phi\omega}\begin{bmatrix} \boldsymbol{p}'_e \\ \boldsymbol{\Phi}'_e \end{bmatrix} = \boldsymbol{J}_g^{-1}\boldsymbol{J}_{\Phi\omega}\boldsymbol{x}'_e \end{aligned} \tag{18}$$

where

$$\boldsymbol{J}_{\Phi\omega} = \begin{bmatrix} \boldsymbol{E}_3 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{N}_{\Phi e} \end{bmatrix}$$

Similar to $\boldsymbol{N}_\phi$ in Eq. (5), $\boldsymbol{N}_{\Phi e}$ is the matrix in the relationship between $\boldsymbol{\omega}_e$ and $\dot{\boldsymbol{\Phi}}_e$. The angular velocity of the base is

$$\boldsymbol{\omega}'_0 = \boldsymbol{J}_{bm\omega}\boldsymbol{\Theta}' \tag{19}$$

The unit quaternion $\boldsymbol{Q}_0$ is employed to represent the attitude of the base. The derivative of $\boldsymbol{Q}_0$ can be written as

$$\begin{bmatrix} q'_0 \\ \boldsymbol{q}' \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -\boldsymbol{q}^{\mathrm{T}} \\ q_0\boldsymbol{E} - \boldsymbol{q}^{\times} \end{bmatrix}\boldsymbol{J}_{bm\omega}\boldsymbol{\Theta}' \tag{20}$$

If the integration step is $\mathrm{d}\tau$, the joint angle is derived as

$$\boldsymbol{\Theta}(\tau + \mathrm{d}\tau) = \boldsymbol{\Theta}(\tau) + \boldsymbol{\Theta}'\mathrm{d}\tau \tag{21}$$

Remark 3. In the process of the trajectory planning, according to Eq. (18), the angular velocity of the end-effector $\boldsymbol{\omega}'_e$ is solved using the derivative of the attitude. In terms of the base, the information of the attitude is obtained based on the angular velocity $\boldsymbol{\omega}'_0$ as shown in Eq. (20). Therefore, based on Remark 1, the Euler angles and unit quaternion are employed to represent the attitude of the end-effector and the base, respectively.

3.2. Avoiding dynamic singularities combining DLS and feedback compensation

If the matrix $\boldsymbol{J}_g$, relating joint velocities to the linear and angular velocities of the end-effector, is not invertible, enormous joint velocities are required according to Eq. (18), which can lead to the failure of trajectory planning. DLS is one of the effective methods to avoid such singularities. The joint velocities are rewritten as

$$\boldsymbol{\Theta}' = \boldsymbol{J}_g^*\boldsymbol{J}_{\Phi\omega}\boldsymbol{x}'_e \tag{22}$$

where $\boldsymbol{J}_g^*$ is damped pseudoinverse corresponding to $\boldsymbol{J}_g$:

$$\boldsymbol{J}_g^* = (\boldsymbol{J}_g^{\mathrm{T}}\boldsymbol{J}_g + \lambda^2\boldsymbol{E}_n)^{-1}\boldsymbol{J}_g^{\mathrm{T}} \tag{23}$$

where λ is called the damping factor. $\boldsymbol{\Theta}'$ obtained through Eq. (22) is the solution to the following DLS problem:

$$\min_{\boldsymbol{\Theta}'}\left(\left\|\boldsymbol{x}'_e - \boldsymbol{J}_g\boldsymbol{\Theta}'\right\|^2 + \lambda^2\left\|\boldsymbol{\Theta}'\right\|^2\right) \tag{24}$$

The damping factor λ is adjusted as follows

$$\lambda^2 = \begin{cases} 0 & \text{if} \quad \hat{\sigma}_6 \geq \varepsilon_6 \\ (1 - (\hat{\sigma}_6/\varepsilon_6)^2)\lambda_0^2 & \text{otherwise} \end{cases} \tag{25}$$

where $\hat{\sigma}_6$ is the estimation of the minimum singular value of the matrix $\boldsymbol{J}_g$, ε_6 defines a threshold which implies the boundary of the neighborhood of singularities, and λ_0 is the maximum value for the damping factor.

Apparently, if DLS is employed to avoid dynamic singularities, the end-effector of the space robot deviates from the desired trajectory. To make up for this deficiency, after the space robot passes the singularities, feedback compensation is introduced. The aim of feedback compensation is that the space robot returns to the desired trajectory after some time.

If $\boldsymbol{x}_{ef}(\tau) = \left[\boldsymbol{p}_{ef}^{\mathrm{T}}(\tau)\quad \boldsymbol{\Phi}_{ef}^{\mathrm{T}}(\tau)\right]^{\mathrm{T}}$ represents the actual trajectory, the joint velocities are

$$\boldsymbol{\Theta}' = \boldsymbol{J}_g^*\boldsymbol{J}_{\Phi\omega}(\boldsymbol{x}'_e + \boldsymbol{u}) \tag{26}$$

where $\boldsymbol{u}$ is termed the feedback compensation, whose initial value is $\boldsymbol{u}(0) = \boldsymbol{0}$. In this paper, the trajectory based on $(\boldsymbol{x}'_e + \boldsymbol{u})$ is called the planned trajectory.

Denote the time thresholds from dynamic-singularity situations to normal situations as τ_i, $i = 1, 2, \ldots$. These points satisfy $\hat{\sigma}_6 < \varepsilon_6$ at $\tau = \tau_i - \mathrm{d}\tau$, and $\hat{\sigma}_6 \geq \varepsilon_6$ at $\tau = \tau_i$. The number of τ_i represents the number of occurrences of singularities.

Define $\tau_0 = 0$. For $\tau \in (\tau_{i-1}\quad \tau_i]$, we suppose that, at the time instant τ_{i-1}, the end-effector state of the planned trajectory is the same as that of the actual trajectory $\boldsymbol{x}_{ef}(\tau_{i-1})$. Then, the error between the planned trajectory and the actual trajectory is

$$\begin{aligned} \boldsymbol{\Delta}_x(\tau) &= \boldsymbol{x}_{ef}(\tau_{i-1}) + \textstyle\int_{\tau_{i-1}}^{\tau}(\boldsymbol{x}'_e + \boldsymbol{u})\mathrm{d}\tau - \boldsymbol{x}_{ef}(\tau) \\ &= \left(\boldsymbol{x}_e(\tau) + \textstyle\int_{\tau_{i-1}}^{\tau}\boldsymbol{u}\mathrm{d}\tau - \boldsymbol{\Delta}_{sum}\right) - \boldsymbol{x}_{ef}(\tau) \end{aligned}$$

where $\boldsymbol{\Delta}_{sum} = \boldsymbol{x}_e(\tau_{i-1}) - \boldsymbol{x}_{ef}(\tau_{i-1})$, which represents the trajectory deviations before τ_{i-1}. Denote the trajectory error corresponding to the i-th neighborhood of singularities as $\boldsymbol{\Delta}_i$. Then $\boldsymbol{\Delta}_i = \boldsymbol{\Delta}_x(\tau_i)$ holds.

Remark 4. τ_i is obtained based on τ_{i-1}. Define a threshold ε_x, which can determine whether the actual trajectory deviates from the planned trajectory. $\tau_i = \tau'$, if τ' satisfies $\hat{\sigma}_6 \geq \varepsilon_6$ and $\left\|\boldsymbol{\Delta}_x(\tau')\right\| > \varepsilon_x$.

$\boldsymbol{u}_i(\tau)$ is designed to reduce or even eliminate the error $\boldsymbol{\Delta}_i$. Therefore, $\boldsymbol{u}_i(\tau)$ satisfies $\int_{\tau_i}^{\infty}\boldsymbol{u}_i(\tau)\mathrm{d}\tau = \boldsymbol{\Delta}_i$. Define $\boldsymbol{u}_0(\tau) = \boldsymbol{0}$, the feedback compensation $\boldsymbol{u}(\tau)$ can be designed as

$$\boldsymbol{u}(\tau) = \sum_{i=0}^{k}\boldsymbol{u}_i(\tau), \quad \tau \in \left(\tau_k\quad \tau_{k+1}\right]$$

To smooth the trajectory, the method of trapezoidal programming is employed to design $\boldsymbol{u}_i(\tau)$:

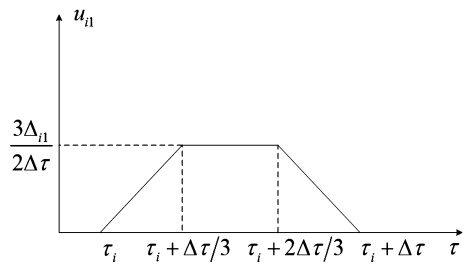

Fig. 2. Schematic diagram of u_{i1}.

$$\boldsymbol{u}_i(\tau) = \begin{cases} \mathbf{0} & \tau \le \tau_i \\ \frac{9\boldsymbol{\Delta}_i}{2\Delta\tau^2}(\tau-\tau_i) & \tau_i < \tau \le \tau_i + \frac{\Delta\tau}{3} \\ \frac{3\boldsymbol{\Delta}_i}{2\Delta\tau} & \tau_i + \frac{\Delta\tau}{3} < \tau \le \tau_i + \frac{2\Delta\tau}{3} \\ \frac{3\boldsymbol{\Delta}_i}{2\Delta\tau} - \frac{9\boldsymbol{\Delta}_i}{2\Delta\tau^2}(\tau-\tau_i-\frac{2}{3}\Delta\tau) & \tau_i + \frac{2\Delta\tau}{3} < \tau \le \tau_i + \Delta(\tau) \\ \mathbf{0} & \tau > \tau_i + \Delta(\tau) \end{cases} \tag{27}$$

where $\Delta\tau$ is a constant to be solved, which represents the time taken to deal with the error τ_i. The first variable of $\boldsymbol{u}_i$ is taken as an example, and it changes with τ as shown in Fig. 2.

According to section 3.1 and Eq. (26), if P_{p2}, $P_{\psi 2}$ and $\Delta\tau$ are determined, the actual trajectory of the end-effector is determined. The flow chart of trajectory planning combining DLS and feedback compensation is shown as Fig. 3.

3.3. Cost function

Because of the nonholonomic constraints, the base attitude is disturbed by the motion of the space robot. This is not expected in certain missions such as communication with the earth. However, for a non-redundant space robot, if the trajectories of the end-effector are given, it is quite challenging to guarantee that the angular velocity of the base is zero according to Eq. (19). Therefore, researchers usually employ optimizations to minimize the base attitude disturbance. In this section, the change of the base attitude from $\boldsymbol{\Phi}_{0s} = [\alpha_{0s} \quad \beta_{0s} \quad \gamma_{0s}]^{\mathrm{T}}$ to $\boldsymbol{\Phi}_{0f} = [\alpha_{0f} \quad \beta_{0f} \quad \gamma_{0f}]^{\mathrm{T}}$ is optimized, and the cost function corresponding to the disturbance is set as

$$J_b = \frac{|\alpha_{0f} - \alpha_{0s}|}{K_\alpha} + \frac{|\beta_{0f} - \beta_{0s}|}{K_\beta} + \frac{|\gamma_{0f} - \gamma_{0s}|}{K_\gamma} \tag{28}$$

where K_α, K_β and K_γ are weighting coefficients.

Bézier curves are employed in this paper, which can limit the joint velocities by adjusting the moving time. Since the normalized time is defined as $\tau = t/T_f$, $\boldsymbol{\Theta}' = T_f\dot{\boldsymbol{\Theta}}$ is obtained. Suppose the joint velocities satisfy the following inequalities:

$$-\dot{\theta}_{i\max} \le \dot{\theta}_i \le \dot{\theta}_{i\max}, \quad i = 1, 2, \ldots, n \tag{29}$$

Then, the moving time is required to satisfy the following inequality:

$$T_f \ge \max\left(\frac{|\theta_i'|_{\max}}{\dot{\theta}_{i\max}}\right) \tag{30}$$

We can now define a cost function related to the moving time as

$$J_t = \frac{T_f}{K_t} \tag{31}$$

where K_t is also a weighting coefficient. The weighting coefficients can be adjusted based on the missions.

Moreover, in Cartesian trajectory planning, it is also challenging to satisfy the limits on the joint angles:

$$-\theta_{i\max} \le \theta_i \le \theta_{i\max} \quad i = 1, 2, \ldots, n \tag{32}$$

For the joint angle requirements, a penalty function is employed, that is

$$J_{penalty} = \sum_{i=1}^{n} P_i \tag{33}$$

where

$$P_i = \begin{cases} 0 & \max|\theta_i| \le \theta_{i\max} \\ 1 & \max|\theta_i| > \theta_{i\max} \end{cases}$$

Finally, the trajectory planning of a space robot is transformed into an optimization problem as follows.

$$\begin{aligned} &\min_{P_{p2}, P_{\psi 2}, \Delta\tau} J = J_b + J_t + J_{penalty} \\ &s.t. \quad \boldsymbol{\Theta}' = \boldsymbol{J}_g^* \boldsymbol{J}_{\Phi\omega}(\boldsymbol{x}_e' + \boldsymbol{u}) \\ &\qquad \begin{bmatrix} q_0' \\ \boldsymbol{q}' \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -\boldsymbol{q}^{\mathrm{T}} \\ q_0\boldsymbol{E} - \boldsymbol{q}^{\times} \end{bmatrix} \boldsymbol{J}_{bm\omega}\boldsymbol{\Theta}' \\ &\qquad \dot{\boldsymbol{p}}_e(t) = \dot{s}_p(t)(\boldsymbol{p}_{ed} - \boldsymbol{p}_{e0}) \\ &\qquad \dot{\boldsymbol{\Phi}}_e(t) = \dot{s}_\Phi(t)(\boldsymbol{\Phi}_{ed} - \boldsymbol{\Phi}_{e0}) \\ &\qquad T_f \ge \max\left(\frac{|\theta_i'|_{\max}}{\dot{\theta}_{i\max}}\right) \end{aligned} \tag{34}$$

4. Chaotic particle swarm optimization

4.1. Particle swarm optimization

Particle swarm optimization (PSO) is applied to many optimization problems due to its simple mechanism. It solves a problem by moving a population of candidate solutions, here called particles, with certain velocities.

Suppose there are M particles, and the position and velocity of the i-th particle are $\boldsymbol{x}_i$ and $\boldsymbol{v}_i$, respectively. The velocity of the i-th particle is influenced by its local best position $\boldsymbol{P}_{li}$ and the global best of the swarm $\boldsymbol{P}_g$ as follows

$$\boldsymbol{v}_i^{k_{iter}+1} = w\boldsymbol{v}_i^{k_{iter}} + c_1 r_1(\boldsymbol{P}_{li} - \boldsymbol{x}_i^{k_{iter}}) + c_2 r_2(\boldsymbol{P}_g - \boldsymbol{x}_i^{k_{iter}})$$

where k_{iter} is the number of the iterations, w is the inertial weight, c_1 and c_2 are the local and global acceleration constants, and r_1 and r_2 are uniformly distributed random numbers. To improve the performance of the algorithm, the inertial weight w is determined based on the form proposed by Musaed [33]. The updated velocity vector is then used to update the position vector of each particle using the following expression:

$$\boldsymbol{x}_i^{k_{iter}+1} = \boldsymbol{x}_i^{k_{iter}} + \boldsymbol{v}_i^{k_{iter}+1}$$

In general, if the maximum number of iterations k_{max} is reached, or $\boldsymbol{P}_g$ is better than the required fitness, the iterations end.

4.2. Chaotic particle swarm optimization

With standard particle swarm optimization, premature phenomenon may occur in some cases, which means that the solution converges to the local optimum. Therefore, chaotic particle swarm optimization (CPSO) is employed in this section. Because of randomness and ergodic statistics property, chaotic sequences can help the particle swarm keep diverse and avoid the premature stop.

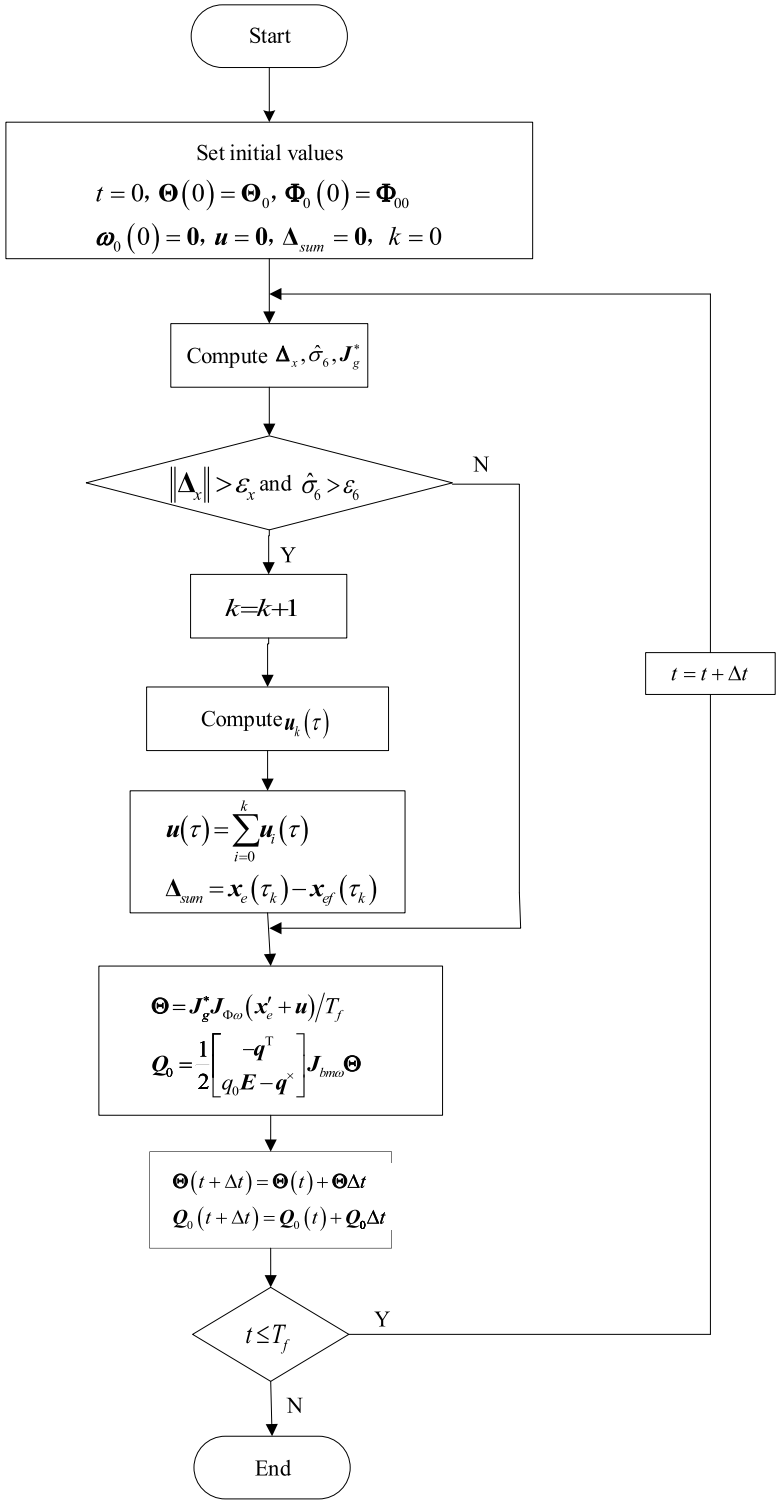

Fig. 3. Flow chart of trajectory planning combining DLS and feedback compensation.

Chaos is a relatively common phenomenon in nonlinear systems. Logistic mapping is a typical chaotic sequence, and the mapping relationship is as follows

$$z_{i+1}=\mu z_i(1-z_i) \quad i=0,1,2,\ldots \tag{35}$$

where μ is the control variable. When $0<z_0<1$ and $\mu=4$, the logistic map is completely chaotic.

The flow chart of CPSO algorithm is shown in Fig. 4. Different from standard PSO, at each iteration, after obtaining the global best position $\boldsymbol{P}_g$, a logistic chaotic algorithm based on Eq. (35) is applied to $\boldsymbol{P}_g$, which is called the local chaotic search of $\boldsymbol{P}_g$. The specific procedures of CPSO algorithm are as follows:

Step 1 Initialize the number of the iterations $k_{iter}=0$, and set the sizes of a population and a particle as M and D, respectively. Produce M particles and their velocities randomly.

Step 2 Calculate the fitness values of particles according to the cost function.

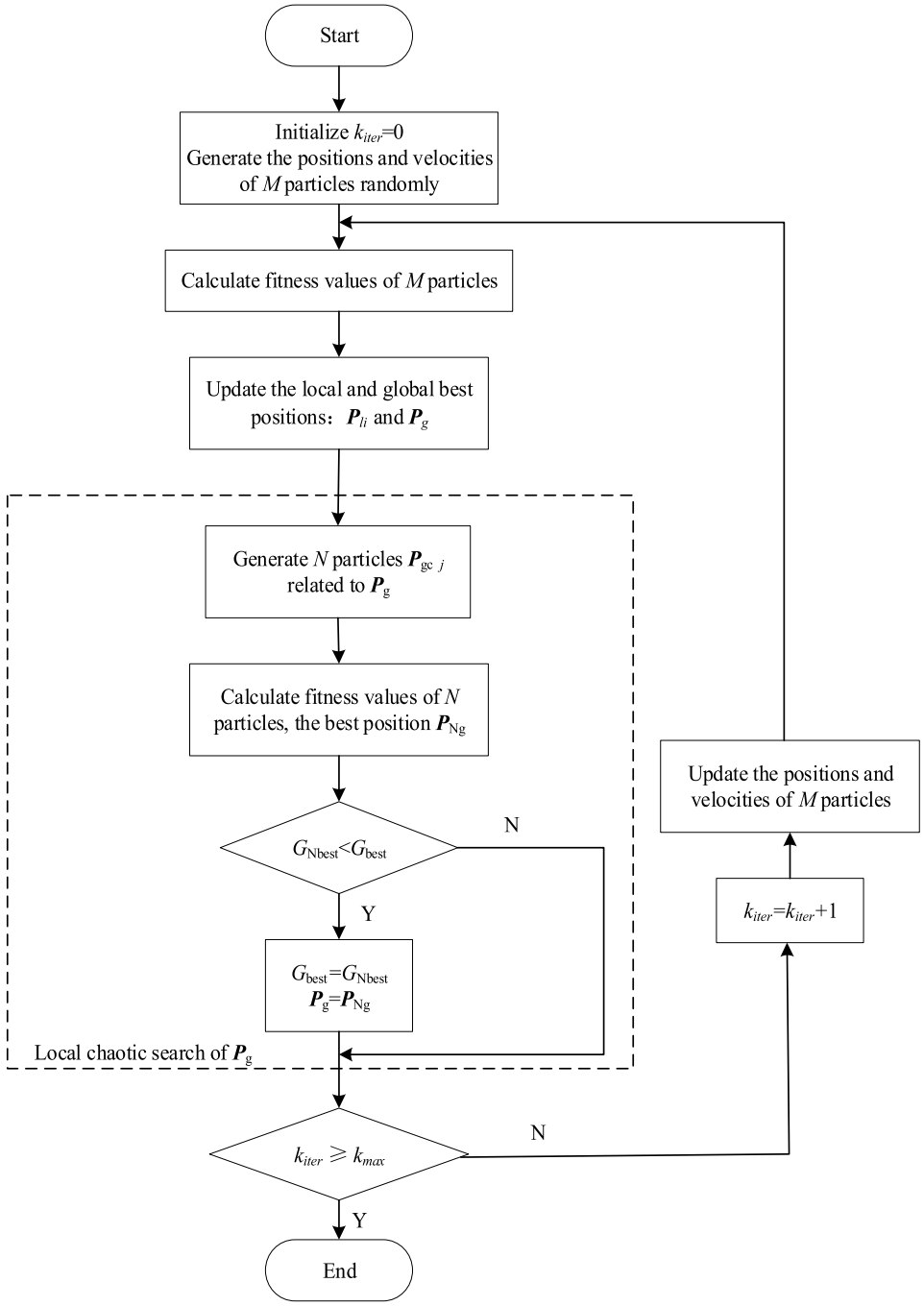

Fig. 4. Flow chart of CPSO algorithm.

Step 3 Update the local best position of i-th particle $\boldsymbol{P}_{l\,i}$ and the global best of the swarm $\boldsymbol{P}_g$. Their fitness values are G_i and G_{best}, respectively.

Step 4 Local chaotic search of $\boldsymbol{P}_g$.

1. A new swarm of N particles is generated related to $\boldsymbol{P}_g$. Firstly, produce a D-dimension vector $\boldsymbol{z}_0$. Then, use Logistic mapping (35) to produce D chaotic sequences $\boldsymbol{Z} = [\boldsymbol{z}_1 \quad \boldsymbol{z}_2 \quad \ldots \quad \boldsymbol{z}_N]$ and N particles $\boldsymbol{P}_{c\,j}(j = 1, 2, \ldots, N)$ as follows:

   $$\boldsymbol{P}_{c\,j} = \boldsymbol{a} + \boldsymbol{z}_j^{\mathrm{T}}(\boldsymbol{b} - \boldsymbol{a})$$

   where $\boldsymbol{a}$ and $\boldsymbol{b}$ represent the lower and upper boundaries of the particles. Finally, N particles of the new swarm $\boldsymbol{P}_{gc\,j}$ are obtained based on $\boldsymbol{P}_g$ and $\boldsymbol{P}_{c\,j}$:

   $$\begin{cases} \boldsymbol{P}_{gc\,j} = (1 - \mu_{k_{iter}})\boldsymbol{P}_g + \mu_{k_{iter}}\boldsymbol{P}_{c\,j} \\ \mu_{k_{iter}} = 1 - ((k_{iter} - 1)/k_{iter})^{\zeta} \end{cases}$$

   where $\mu_{k_{iter}}$ is the factor corresponding to $\boldsymbol{P}_{c\,j}$, and ζ is used to adjust $\mu_{k_{iter}}$.
2. Calculate the fitness values by substituting N particles into the cost function. The best position of N particles is $\boldsymbol{P}_{Ng}$ and its fitness value is G_{Nbest}. If $G_{Nbest} < G_{best}$ holds, then the original global best $\boldsymbol{P}_g$ is replaced by $\boldsymbol{P}_{Ng}$, that is $\boldsymbol{P}_g = \boldsymbol{P}_{Ng}$.

Step 5 Update the positions and velocities of M particles.

Step 6 Check whether the stop conditions are met. If so, the process of CPSO ends and outputs the best position $\boldsymbol{P}_g$. If not, the procedures return to step 2.

Table 2
Mass and inertial parameters of space robot.

		B_0	B_1	B_2	B_3	B_4	B_5	B_6
Mass (kg)		400	6	5	5	4	3	2
$\boldsymbol{a}_i$ (m)			0	0.2702	0	0	0	0
			0	0	0	0	-0.0338	0
			0.15	-0.2513	0.15	-0.35	0	0.0750
$\boldsymbol{b}_i$ (m)		0.3570	0	0.5598	0	0	0	0
		-0.0095	0	0	0	0	-0.0662	0
		0.419	0.15	-0.0487	0.15	-0.35	0	0.1595
$\boldsymbol{I}_i$ (kg · m^2)	I_{xx}	30	0.15	0.0926	0.105	0.2498	0.0330	0.05152
	I_{yy}	28	0.15	0.9053	0.105	0.2498	0.0172	0.05152
	I_{zz}	32	0.075	0.8451	0.0294	0.0196	0.0260	0.02192
	I_{xy}	0.26	0	0	0	0	0	0
	I_{xz}	0.37	0	0.1315	0	0	0	0
	I_{yz}	-0.29	0	0	0	0	0	0

Table 3
Ranges of joint angles.

i	$[\theta_{imin} \quad \theta_{imax}]$ (°)
1	[−150 150]
2	[−150 150]
3	[−180 180]
4	[−150 150]
5	[−90 90]
6	[−180 180]

In general, when the initial and desired states of the space robot are obtained, the ground teleoperation system solves the optimal trajectory using CPSO. Then, these data related to the above trajectory will be sent to the space robot system by uplink channel. Finally, the space robot is controlled to move along the optimal trajectory.

5. Simulation results

The space robotic system in this section is composed of a spacecraft and a 6 DOF manipulator as shown in Fig. 1. The mass and inertia properties of the space robot are listed in Table 2. The ranges of the joint angles are shown in Table 3, and the joint velocities satisfy $|\dot{\theta}_i| \le \dot{\theta}_{\text{imax}} = 5°/s$.

The initial position and attitude of the end-effector are

$$\boldsymbol{p}_{e0} = [1.9495 \quad -0.0137 \quad 1.3940]^{\mathrm{T}} \text{ (m)}$$
$$\boldsymbol{\Phi}_{e0} = [180 \quad -82 \quad 0]^{\mathrm{T}} \text{ (°)}$$

The corresponding configuration of joints and the initial attitude of the spacecraft are

$$\boldsymbol{\Theta}_0 = [0 \quad 40 \quad -27 \quad 0 \quad -5 \quad 0]^{\mathrm{T}} \text{ (°)}$$
$$\boldsymbol{\Phi}_{00} = [0 \quad 0 \quad 0]^{\mathrm{T}} \text{ (°)}$$

The desired position and attitude of the end-effector are listed as follows

$$\boldsymbol{p}_{ed} = [1.7185 \quad 0.1041 \quad 1.7335]^{\mathrm{T}} \text{ (m)}$$
$$\boldsymbol{\Phi}_{ed} = [155.31 \quad -58.81 \quad 28.33]^{\mathrm{T}} \text{ (°)}$$

The optimal problem (34) is solved with PSO and CPSO respectively. As shown in Table 4, although the changes of the spacecraft attitude with these two methods are really similar, the moving time with CPSO is much less than that with PSO. The best fitness values G_{best} change with iterations as shown in Fig. 5. According to Fig. 5, at the second iteration, G_{best} obtained with PSO converges

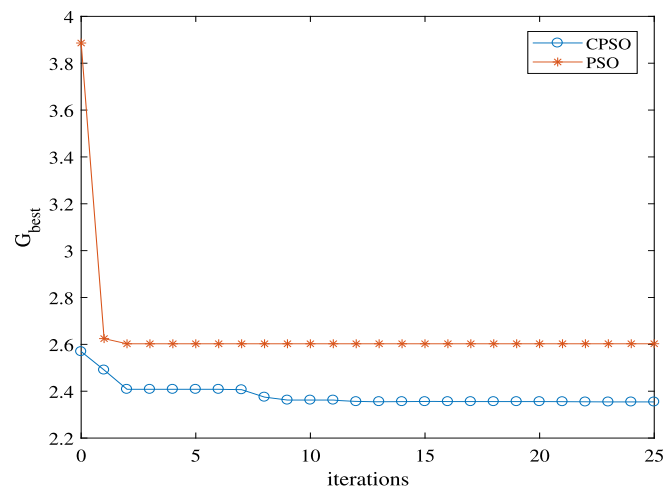

Fig. 5. Changes of the best fitness values.

to a local optimal solution, which is considered as the final solution to this problem. It means that the premature phenomenon occurs. And the initial G_{best} with CPSO is less than the final G_{best} with PSO. Therefore, based on Table 4 and Fig. 5, with CPSO, the solution is better and the premature is avoided effectively.

The optimal solutions with CPSO are substituted into Eq. (13) and Eq. (27). The corresponding joint angles and velocities are obtained as Fig. 6 and Fig. 7, respectively. The joints move smoothly and satisfy the requirements in Table 3 according to Fig. 6. In Fig. 7, the joint velocities are no more than $5°/s$.

Avoiding dynamic singularities combining DLS and feedback compensation has been proposed in this paper. The method is here compared with DLS without feedback compensation. Taking the Y-axis of the position as an example, the actual trajectories of the end-effector and the errors with these two methods are shown in Fig. 8 and Fig. 9 respectively. According to Fig. 9, the space robot system meets the dynamic singularities from 4.013 s to 9.788 s. As a consequence, the trajectories of the end-effector deviate from the desired trajectory. With DLS, the trajectory continues to deviate from the desired trajectory. On the other hand, with the proposed method, the trajectory errors reduce gradually and the space robot returns to the desired trajectory after 14.230 s. Therefore, the proposed method can reduce the trajectory errors caused by DLS.

The base attitude disturbance and moving time are considered to optimize the trajectory in this paper. If the parameters m and $\Delta\tau$ are set to 3 and 0.176, a trajectory for the end-effector is generated, which can be termed the trajectory without optimiza-

Table 4
Optimal results of CPSO and PSO.

	CPSO	PSO
Best fitness value	2.354	2.602
Change of base attitude (°)	$[-2.5475 \quad 3.1580 \quad 2.0828]^T$	$[-2.5754 \quad 3.1122 \quad 1.9728]^T$
Moving time (s)	23.89	32.11

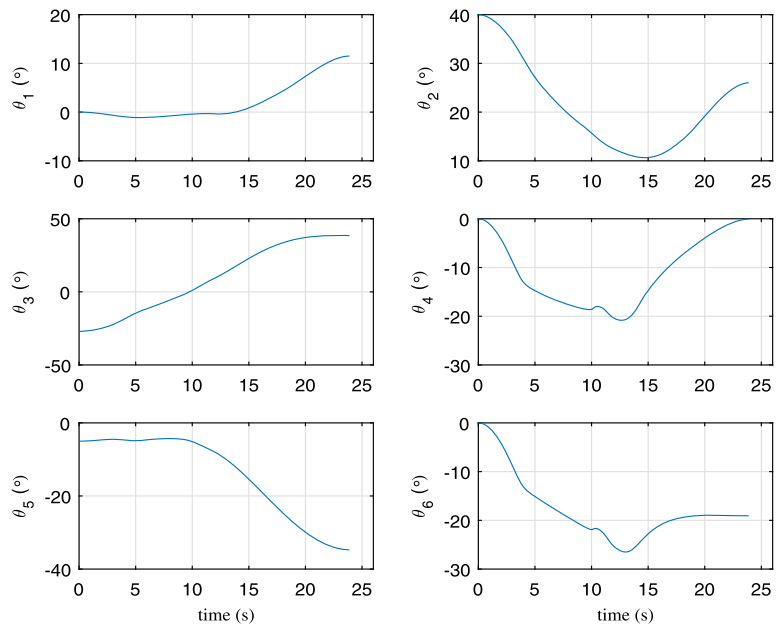

Fig. 6. Simulation results of joint angles.

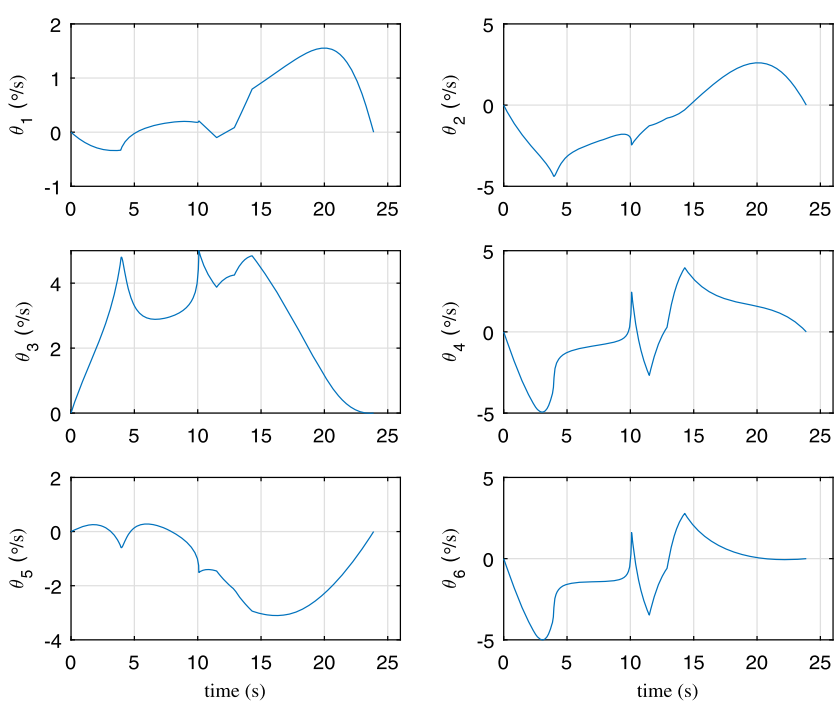

Fig. 7. Simulation results of joint velocities.

tions. The optimized trajectory and the trajectory without optimizations are compared here. The base attitude disturbances $\mathbf{\Phi}_e$ represented in Euler angles (ZYX) are shown in Fig. 10. The final base attitude disturbances are $[-2.5475° \quad 3.1580° \quad 2.0828°]^T$ and $[-1.644° \quad 14.340° \quad 0.7124°]^T$, whose corresponding rotation angles are 4.5850° and 14.4628°, respectively. The moving time of the former (23.89 s) is less than that of the latter (25.80 s). Therefore, in terms of both base attitude disturbance and moving time, the former is better than the latter. This demonstrates that the optimization of the trajectories is effective.

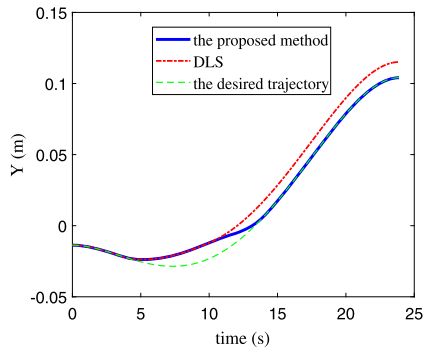

Fig. 8. Trajectories of the proposed method and DLS.

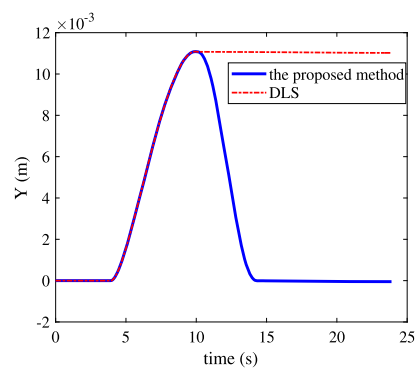

Fig. 9. Errors of the proposed method and DLS.

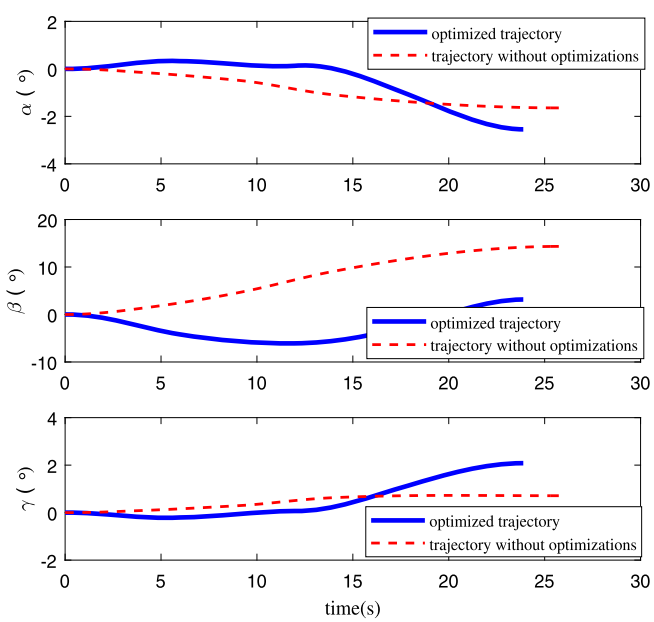

Fig. 10. The base attitude disturbance in Euler angles (ZYX).

6. Conclusions

This paper studied Cartesian trajectory planning for space robot systems. Dynamic singularities were avoided by combining DLS and feedback compensation. Considering the base attitude disturbance and the moving time, the trajectory planning was transformed into an optimization problem and CPSO algorithm was used to solve the optimization. Moreover, because of simplicity and normalization, Bézier curve was employed to parametrize the trajectories of the end-effector. The simulation results demonstrated the effectiveness of the developed trajectory planning algorithm. CPSO can improve the premature phenomenon of the standard PSO. Compared with DLS, the proposed method can reduce the trajectory errors gradually. Compared with the trajectory without optimizations, in terms of both moving time and base attitude disturbance, the performances of the optimized trajectory are better. Besides, the constraints on the joint velocities and positions are also guaranteed with the proposed method.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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